MILITARY SPECIFICATION

BATTERY, SEA WATER

This amendment forms a part of Military Specification MIL-B-82439A(OS) dated 3 July 1969.

Wherever it appears in the specification, delete "Bureau of Naval Weapons" and substitute "Naval Sea Systems Command"; also delete "BuWeps" and substitute "NAVSEA".

*Page 2, Paragraph 2.1. Under "DRAWINGS" add the following:

"Dwg 2526791	Wire, Arming
Dwg 2526889	Sleeve
Dwg 2574743	(Alternate Configuration) Case, Battery
Dwg 2574740	(Alternate Configuration) Battery Cell Assembly"

*Page 4, paragraph 3.3.3. Add following sentence to end of paragraph: "There shall be no separation of the arming wire from the sleeve when subjected to a pull test of 175 +10 -0 pounds".

Page 8, paragraph 4.3.1, line 3. Delete "4.4 through 4.4.4.15" and substitute "4.4 through 4.4.4.17'!.

Page 9, Table IV. Delete and substitute new Table IV (see page 2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (Code 501), Indian Head, MD 20640 by using the self-addressed Standardization Documentation Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

"TABLE IV

First Article Test Sequence

Tests	Paragraph Reference	Test Sequence							
		A	B	C	D	E	F	G	H
Inspection	4.4.4.1	1	1	1	1	1	1	1	1
Vibration, Ready Condition	4.4.4.2	2							
Acceleration, Read Condition	4.4.4.3		2						
Shock, Ready Condition	4*4.4*4		3						
Temperature and Humidity, Non-Operating Condition	4.4.4.5	3		2					
Marine Atmosphere, Ready Condition	4.4.4.6						2		
Vibration, Packed and Packaged	4.4.4.7				2				
Shock, Packed and Packaged	4.4.4.8				3				
High Pressure, External	4.4.4.9[1/]					2	3		
Arming Wire Pull Test	4.4.4.11[1/]	4	4	3	4			3	3
Insulation Resistance, Pin to Case	4.4.4,16	5	5	4	5	3	4	4	4
Circuit Resistance, Pin to Pin	4.4.4.17	6	6	5	6	4	5	5	5
Internal Leakage	4.4.4.10	7	7	6	7	5	6	6	6
Accelerated Storage Life, Packed and Packaged Condition	4.4.4.12							2	
Accelerated Storage Life, Ready Condition	4.4.4.13								2
Low Temperature, Low Salinity Operation	4.4.4.14	8	8			8		7	7
High Temperature, High Salinity Operation	4.4.4615			7		6	7		

[1/]Battery assemblies shall be subjected to the internal leakage requirements of 4.4.4.10 within a 6 hour period, and operated in accordance with 4.4.4.14 or 4.4.4.15, as applicable, within a 48 hour period following the arming wire pull test or removal of the arming wire."

Page 11, paragraph 4.4.4.1. Delete and substitute:

"4.4.4.1 Inspection. The battery assemblies shall be subjected to the inspections detailed in 4.5.7.2, 4.5.7.6(a)(1), (a)(2), (a)(4), (a)(5), and (a)(6), and 4.5.7.6(b), in-the sequence noted."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

Page 13, paragraph 4.4.4.5: Delete last sentence ("Thereafter, 4 battery assemblies shall be...").

* Page 14, paragraph 4.4.4.11. Delete and substitute:
"4.4.4.11 Arming wire pull test. The battery assembly shall be subjected to test (a), test (b), and test (c). The battery assembly shall be readied for test (a) and thereafter test (b) by cutting through the 22-1/4 inch leg of the arming wire in the red cutoff area near the 11/32 inch x 45° wire bend."

* Page 15. After paragraph 4.4.4.11(b) add following new paragraph:
"(c) A random sample of two arming wires out of each production lot (150 units maximum) shall be cut at a convenient distance from the sleeve and removed from the battery assembly. The 1/2 inch diameter and 1/4 inch diameter eyes of each arming wire shall be individually subjected to a pull test of 175 +10 -0 pounds and held for one minute. The eye is allowed to deform, but separation of the wire from the sleeve shall be cause for rejection of the arming wire production lot."

Page 15, paragraphs 4.4.4.12 and 4.4.4.13. Delete and substitute:
"4.4.4.12 Accelerated storage life, packed and packaged condition. The battery assembly, in the packed and packaged condition, shall be subjected to an environmental temperature of +140°F +50 -0°F, at ambient atmospheric conditions, for a period of 30 days.

4.4.4.13 Accelerated storage life, ready condition. The battery assembly, in the ready condition, shall be subjected to an environmental temperature of +140°F +5° -0°F, at ambient atmospheric conditions for a period of 14 days."

Page 15, paragraph 4.4.4.14, line 5: Delete "The test fixture and electrolyte, Type A," and substitute "The electrolyte, Type A,".

Page 15, paragraph 4.4.4.15, lines 5 and 6: Delete "The test fixture and electrolyte, Type B," and substitute "The electrolyte, Type B,".

Page 15. Add the following new paragraphs:
"4.4.4.16 Insulation resistance, pin-to-case. With the arming wire removed, the resistance between both pins A and B of the battery connector and the battery case shall exceed 20 megohms when measured at a potential of 50 ±10 volts direct current.

4.4.4.17 Circuit resistance, pin-to-pin. With the arming wire removed, the resistance between pins A and B of the battery connector shall be 470 K ±94 K ohms."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

Page 16, paragraph 4.4.5: In line 2 delete "by the supplier" and substitute "by the testing facility"; under "Battery Test Data" delete entries (g) and (h) and substitute:

"(g) Resistance
pin-to-pin (Arming Wire Installed) _____
pin-to-case (Arming Wire Removed) _____
pin-to-pin (Arming Wire Removed) _____

(h) Operation Test, Low Temp. _____ High Temp. _____
Battery Assembly _____ Degrees F
Electrolyte _____ Degrees F
Electrolyte _____ Percent Salinity
output
Time in seconds:
0.1, 0.2, 0.3, 0.4, 0.5, 1.0, 2.0, 3.0, 5.0, 10.0
Voltage _ . _ _ _ _ _ _ _ _ _ ____

(i) Polarity: Pin A _____, Pin B _____

(j) Configuration _____"

Page 16, paragraph 4.4.6: Delete "through 4.4.4.15" and substitute "through 4.4.4(17".

* Page 19, paragraph 4.5.7.6. Delete paragraph title and substitute: "4.5.7.6 Configuration (Refer to Dwg 2526883 or 2574743 (alternate configuration))"; also add "(7) 2.121 +.002, -.004 DIA."

* Page 19, paragraph 4.5.7.7. Delete and substitute: "4.5.7.7 Arming wire pull test. The battery assembly shall be subjected to test (a), test (b), and test (c). The battery assembly shall be readied for test (a) and thereafter test (b) by cutting through the 22-1/4 inch leg of the arming wire in the red cutoff area near the 11/32 inch x 450 wire bend."

* Page 19. After paragraph 4.5.7.7(b) add the following: "(c) A random sample of two arming wires out of each production lot (150 units maximum) shall be cut at a convenient distance from the sleeve and removed from the battery assembly. The 1/2 inch diameter and 1/4 inch diameter eyes of each arming wire shall be individually subjected to a pull test of 175 +10 -0 pounds and held for one minute. The eye is allowed to deform, but separation of the wire from the sleeve shall be cause for rejection of the arming wire production lot."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

Page 20, paragraph 4.5.8.2: Delete "Low temperature, low salinity operation. The battery cell, after being conditioned for 1 hour minimum at +2°F ±2°F., shall be mounted in a test fixture similar to that shown in Figure 7. The electrical test circuit shall be in accordance with Figure 6. The test fixture -------- ." Substitute "Low temperature, low salinity operation. The 2571089 battery cell shall be mounted in the holding fixture shown in Figure 7, and the 2574740 battery cell assembly (alternate configuration) shall be mounted in a fixture equivalent to the battery case. The electrical test circuit shall be in accordance with Figure 6. After being conditioned for 1 hour minimum at 2°F ±2°F, the battery cell shall be installed in a test fixture similar to that shown in Figure 5. The Test fixture -------- ".

Page 21, paragraph 4.5.9(h): Delete "Arming wire pull test _____" and substitute "Arming wire pull test _____ lbs ."

* Pages 22 and 23. Delete Table IX and substitute:

"TABLE IX

Periodic Test Sequence

Tests	Sample and Test Sequence					
	Sample Lot No.	1, 4, etc.		2, 5, etc.		3, 6, etc.	
	No. of Units	1	2	1	2	1	2
Vibration, Ready Condition					1 1/		1 1/
Acceleration, Ready Condition							2 1/
Shock, Ready Condition			1				3 1/
Temperature and Humidity			2		2		4
Marine Atmosphere			3				
Arming Wire Pull Test (c)			4		3		5
Accelerated Storage Life Test, Ready Condition 2/		1		1		1	
High Pressure, External			5		4		6
Insulation Resistance Pin-to-Pin			6		5		7
Insulation Resistance, Pin-to-Case			7		6		8
Circuit Resistance, Pin-to-Pin			8		7		9
Internal Leakage			9		8		10

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

TABLE IX

Periodic Test Sequence (Cont'd)

Tests	Sample and Test Sequence					
	Sample Lot No.	1, 4, etc.		2, 5, etc.		3, 6, etc.	
	No. of Units	1	2	1	2	1	2
Low Temperature, Low Salinity Operation	2	10	2		2	11
High Temperature, High Salinity Operation				9		

1/Visual check after testing.
2/Battery Assemblies shall be subjected to the Internal Leakage requirements of 4.5.7.5 within a 6 hour period, and operated in accordance with 4.6.3.8 or 4.6.3.9, as applicable, within a 48 hour period following arming wire removal. "

Page 23, paragraph 4.6.3.1. Delete and substitute:
"4.6.3.1 High pressure, external. External high pressure testing of the battery assembly shall be performed in accordance with 4.4.4.9."

* Page 24. Delete paragraphs 4.6.3.7 through 4.6.3.9 and substitute:
"4.6.3.7 The arming wire pull test, subsection (c), shall be performed in accordance with 4.5.7.7(c).

4.6.3.8 Accelerated storage life, ready condition. Accelerated storage life testing of the battery assembly, in the ready condition, shall be performed in accordance with 4.4.4.13.

4.6.3.9 Low temperature, low salinity operation. Low temperature, low salinity operation of the battery assembly shall be performed in accordance with 4.4.4.14.

4.6.3.10 High temperature, high salinity operation. High temperature, high salinity operation of the battery assembly shall be performed in accordance with 4.4.4.15.

4.6.3.11 Insulation resistance, pin-to-pin. With the arming wire installed, perform a pin-to-pin insulation resistance test in accordance with 4.5.7.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:08Z
Check the source to verify that this is the current version before use.

4.6.3.12 Insulation resistance, pin-to-case. With the arming wire removed, perform a pin-to-case insulation resistance test in accordance with 4.4..4.16.

4.6.3.13 Circuit resistance, pin-to-pin. Pin-to-pin circuit resistance testing of the battery assembly shall be performed in accordance with 4.4.4.17.

4.6.3.14 internal leakage. The battery assembly shall be subjected to the internal leakage test in accordance with 4.4.4.10."

* Page 24, paragraph 4.6.5: Delete "4.6.3.1 through 4.6.3.9" and substitute "4.6.3.1 through 4.6.3.14".

Page 25, paragraph 6.3.9: Delete first sentence and substitute "The electrolyte shall consist of tap water and sea salts or sea water adjusted to meet the requirements of Type A and B solutions, as required."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian
Navy - OS

Preparing Activity
Navy - OS
Project No. 1355-N395

Source: https://assist.dla.mil -- Downloaded: 202610-23T07:08Z
Check the source to verify that this is the current version before use.